

operation, results in a GPU temperature of 70.5°C but raises the CPU temperature to 83°C, the highest among all modes. In Quiet mode, the GPU has the lowest temperature at 66.6°C, while the CPU remains at 76°C, striking a balance between noise reduction and thermal management. At idle in Quiet mode, the temperatures drop significantly, with the GPU at 33.5°C and the CPU at 49°C, demonstrating effective cooling when the system is not under load. Overall, the Entertainment mode generates the highest CPU temperatures, while the Quiet mode provides the best balance between noise and heat management.

The figures mentioned above are very impressive. Mainly because the laptop CPU temperatures usually lie in the 90°C range. So the G6X is easily beating out its competitors, at least in terms of cooling.

The maximised cooling also translates to increased performance as the RTX 4050 performs as well as the RTX 4050 in the much more premium ASUS ROG Strix G16 and a lot better than the RTX 4050 in the Lenovo LOQ. Now, provided that the LOQ comes with an AMD Ryzen 7 7840HS, which isn't as powerful as the i3650HX but still. The enhanced cooling of the Gigabyte G6X is to be applauded here for this result.

When it comes to real-world gaming performance, the Gigabyte G6X impresses with one of the highest scores achieved by an RTX 4050 laptop. The performance analysis of three laptops equipped with RTX 4050 graphics cards and various Intel processors—Gigabyte G6X (i3650HX), HP Victus 15 (i3420H), and ASUS ROG Strix G16 (i3650HX)—reveals distinct differences in gaming performance across several popular titles at 1080p resolution on both high and medium settings.

Metro Exodus shows that the HP Victus 15 leads at 1080p high settings with an average FPS of 93, followed by the ASUS ROG Strix G16 at 85 FPS and the Gigabyte G6X at 83 FPS. However, at 1080p medium settings, the ASUS ROG Strix G16 takes the lead with 115 FPS, whereas the Gigabyte G6X and HP Victus 15 score 111 FPS and 71 FPS, respectively.

In Dirt 5, the ASUS ROG Strix G16 consistently performs best, with 106 FPS at high settings and 130 FPS at medium

settings. The Gigabyte G6X follows closely with 97.5 FPS at high settings and 127 FPS at medium settings. The HP Victus 15 lags behind with 85 FPS at high settings and 109 FPS at medium settings.

Assassin's Creed Valhalla results indicate a close competition between the Gigabyte G6X and ASUS ROG Strix G16, both achieving 119 FPS at medium settings. At high settings, the ASUS ROG Strix G16 slightly edges out with 100 FPS, while the Gigabyte G6X and HP Victus 15 score 97 FPS and 91 FPS, respectively. For Shadow of the Tomb Raider, the Gigabyte G6X and ASUS ROG Strix G16 again show similar performance, with the ASUS ROG Strix G16 slightly ahead at both settings—129 FPS at high and 135 FPS at medium, compared to the Gigabyte G6X's 131 FPS at high and 136 FPS at medium. The HP Victus 15 scores lower, with 103 FPS at high and 108 FPS at medium settings.

In Civilization VI, the ASUS ROG Strix G16 leads with 122.91 FPS at high settings and 132.45 FPS at medium settings. The Gigabyte G6X follows with 119 FPS at high and 129 FPS at medium settings, while the HP Victus 15 scores 79 FPS and 83 FPS, respectively. Gears of War 5 shows the ASUS ROG Strix G16 outperforming with 134 FPS at high settings and 163 FPS at medium settings. The Gigabyte G6X trails with 125 FPS at high and 152 FPS at medium settings, and the HP Victus 15 scores 106 FPS and 127 FPS, respectively.

Lastly, GTA V benchmarks highlight the ASUS ROG Strix G16 as the top performer with 183 FPS at high settings and 187 FPS at medium settings. The Gigabyte G6X follows with 167 FPS at high and 171 FPS at medium settings, while the HP Victus 15 scores 143 FPS at high and 156 FPS at medium settings.

Overall, the ASUS ROG Strix G16 consistently delivers the highest FPS across most games and settings, making it the best choice for gamers seeking top-tier performance. The Gigabyte G6X offers solid performance, often close to the ASUS ROG Strix G16, while the HP Victus 15, despite being equipped with the same RTX 4050 GPU, generally falls behind in FPS, especially at medium settings.

VERDICT

In conclusion, the Gigabyte G6X 9MG proves to be a standout option for gamers looking to strike a balance between performance and affordability. While Gigabyte is a well-known name among PC enthusiasts for its quality components, the G6X series marks a solid entry into the laptop gaming market. With the RTX 4050 GPU and Intel i3650HX processor, the G6X 9MG delivers impressive gaming performance, often matching or coming close to the more expensive ASUS ROG Strix G16 and outpacing the HP Victus 15 in several benchmarks.

Despite some drawbacks, such as the tendency to attract fingerprints and a display that lacks the vibrancy of competitors like the Lenovo LOQ, the G6X 9MG shines with its robust build quality, extensive I/O port selection, effective cooling system, and competitive gaming performance. The upgradability options for RAM and storage further enhance its appeal, making it a future-proof choice for both gaming and productivity.

Priced at Rs 96,999, the Gigabyte G6X 9MG offers exceptional value, delivering near top-tier performance without the hefty price tag of the ASUS ROG Strix G16, which costs Rs 1.25 lakh. For gamers on a budget, the Gigabyte G6X 9MG is definitely worth considering, providing a solid mix of performance, durability, and upgradability that makes it a compelling choice in the competitive gaming laptop market.

—Yetnesh Dubey

SPECIFICATIONS

CPU: Intel Core i7-13650HX Processor (24M Cache, up to 4.9 GHz, 14 cores: 6 P-core and 8 E-core) | GPU: NVIDIA GeForce RTX 4050 Laptop GPU 6GB GDDR6 (Boost Clock 2370MHz / Maximum Graphics Power 105 W) | Display: 16.0" 16:10 WUXGA (1920×1200) 165Hz Display (MUX Switch) | Storage: 2x M.2 SSD slots (Supports 2x PCIe Gen4x4 NVMe™ M.2 SSD, Up to 4TB) | RAM: 2x DDR5 Slots (DDR5-4800MHz, Up to 64GB) | Weight: 2.5 kg | Battery Size: Lithium-Ion 73Wh

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